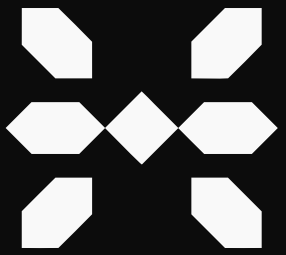
HOOX · AXIS · DOCUMENTATION



PLATFORM ENGINEERS · DEPLOYERS

DEVOPS MANUAL

Local dev, build/serve, Cloudflare, VPS, CORS, CI

Open charting PWA — own the axes, swap the engine

CONTENTS

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OVERVIEW

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title: "DevOps" description: "Build, serve, Cloudflare®, VPS demo, CORS, and CI for the AXIS PWA and Worker."

DEVOPS

Abstract

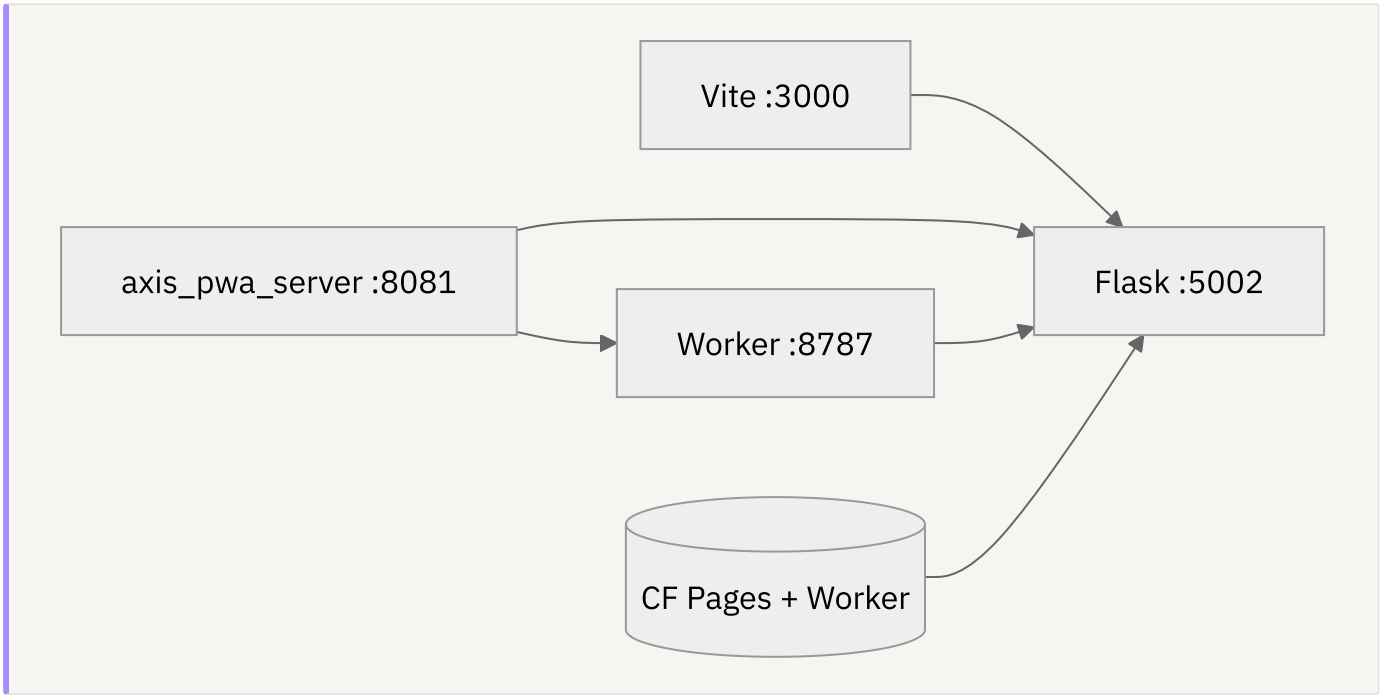
AXIS DevOps spans five runnable surfaces:

1. **Vite + Solid PWA** — product UI (`bun run dev`)
2. **Static server** (`axis_pwa_server.py` or Vite preview) — production-like assets
3. **Cloudflare® Worker** (`worker/`) — edge API / WS / OAuth / on-chain proxy
4. **Desktop (Tauri 2)** — native shell around the same UI (`bun run desktop:dev`)
5. **AXIS CLI** (`packages/cli`) — install, doctor, setup, secrets, deploy, health

See Desktop (Tauri) and AXIS CLI for operator entry points.

Python evaluation for desk workflows still often uses the **Flask Pro API** (`make run` on `:5002`). The Worker proxies there when `EXTERNAL_BACKEND` is set.

Conceptual model



Topology cheat sheet

Mode	Command sketch	Ports
Dev AXIS	`bun run dev` / `bun run axis dev`	Vite (often `:3000`)
Dev + Flask	+ `make -C ../pyne run` (checkout sometimes named `pynescript`)	`:5002`
Static dist	`bun run build` + `python axis_pwa_server.py`	`:8081`
Worker local	`bun run axis dev worker` / `cd worker && bun run dev`	`:8787`
CF Worker prod	`axis deploy`	HTTPS workers.dev / custom
Operator	`axis doctor` · `axis setup` · `axis health`	—

Page map

Page	Topic
Local dev	Day-to-day loop
AXIS CLI	Install, doctor, setup, secrets, deploy, health
Build and serve	Vite build, SPA server, assets
Cloudflare®	Pages + Worker deploy
Desktop (Tauri)	Native shell
VPS demo	Single-box static + Flask
CORS and origins	Allowed origins matrix
CI and testing	GitHub Actions, coverage, e2e
OKF bundle	Compiled repo map, refreshed on commit

Frozen names

Surface	Value
Pages project	`axis`
Pages URL	`https://axis.hoox.sh`
Worker script	`worker-axis`
Worker URL	`https://worker.axis.hoox.sh`
Health service	`worker-axis`

See also

- Worker
- Testing reference
- AXIS hub

LOCAL DEV

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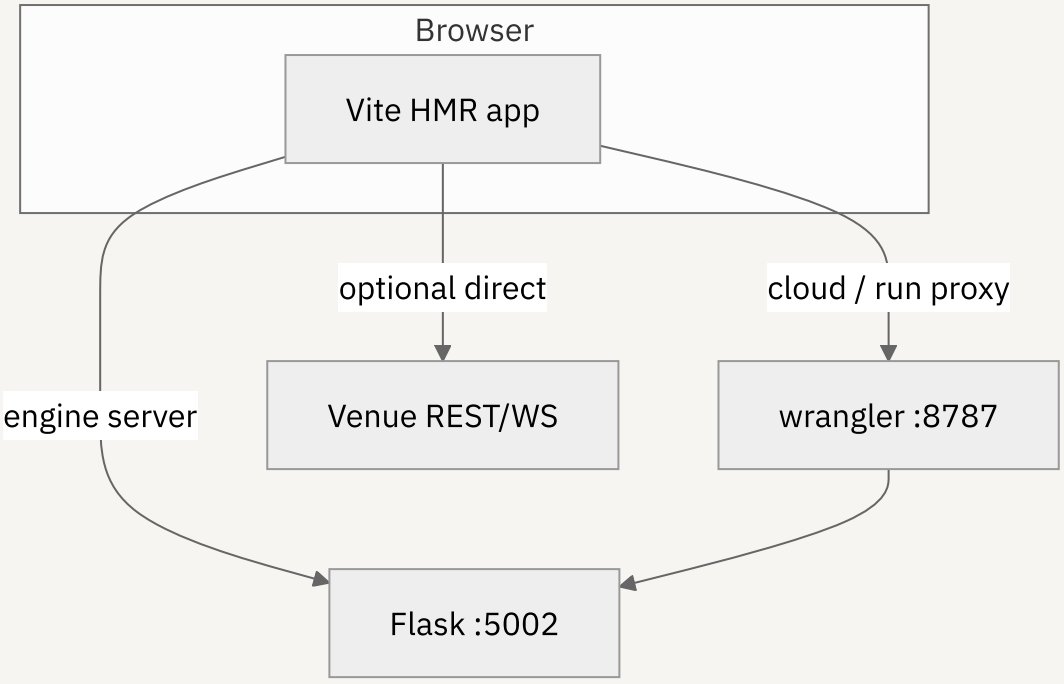
title: "Local development" description: "Day-to-day AXIS loop: Vite, Flask, Worker wrangler, endpoints, and plugin examples."

LOCAL DEVELOPMENT

Abstract

Local AXIS development is a **multi-process** topology. AXIS is a Solid app under Vite; evaluation usually hits Flask; optional cloud/storage/stream features hit the Worker.

Conceptual model



Interface surface — commands

Bootstrap (CLI-first)

```
axis install   # app + worker + CLI
axis doctor    # missing worker/wrangler.toml is a warning until `axis setup`
```

Frontend (preferred product path)

```
bun run dev            # Vite + Solid → typically http://127.0.0.1:3000
# or: bun run axis dev
```

Desktop (optional Tauri shell)

```
bun run desktop:dev    # Vite + native window (see devops/desktop)
# or: bun run axis dev desktop
```

Flask Pro API (engine backend)

From the pyne checkout (sister repo; clone hoox-sh/pyne as `../pyne` — the local directory is sometimes still named `pynescript`):

```
make -C ../pyne run   # Flask → :5002
```

Set PWA endpoint to `http://127.0.0.1:5002` for the **server** engine. Workers Manager and Settings can probe this URL.

Worker

```
bun run axis dev worker   # http://127.0.0.1:8787
# or: cd worker && bun run dev
```

Point cloud storage / Worker endpoint at `http://127.0.0.1:8787`. For `/api/run` proxy, set Worker `EXTERNAL_BACKEND=http://127.0.0.1:5002` (works on local wrangler; not on Cloudflare® edge).

AXIS CLI

Preferred operator entry for install / doctor / setup / deploy — see AXIS CLI:

```
bun run axis install
bun run axis doctor
bun run axis setup
bun run axis dev worker
```

Makefile helpers

Target	Intent
`python axis_pwa_server.py`	Serve built `dist/` on `:8081` (not Vite)
`make worker-dev`	Wrangler dev
`make worker-install`	Install worker deps
`bun run test:unit`	Bun unit tests under `tests/`

Use `bun run dev` for the product, or `bun run build` and serve `dist/`. The pre-Solid `server.ts` static tree was removed in 2.21.0 (legacy shell).

Storybook (component lab)

```
bun run storybook        # isolated UI on :6006 (`storybook-solidjs-vite`)
bun run build-storybook  # static build
```

Stories live in `src/stories/` (topbar fields, theme tokens, studio primitives, draw tools, toasts, …). Covers chrome in isolation — no Flask/Worker needed.

Internals — useful paths

Path	Role
`src/index.tsx`	App entry
`vite.config.*`	Vite / Solid plugin
`src/engines/catalog.ts`	Default engine endpoint
`worker/wrangler.toml`	Local vars
`public/plugins/`	Example modules served at `/plugins/`

Worked loop

1. Terminal A: `make run` (Flask).
2. Terminal B: `bun run dev`.
3. Open Vite URL; set engine **Server-Side**, endpoint `http://127.0.0.1:5002`.
4. Load `mock-walk` if venues are blocked.
5. Optional Terminal C: Worker with `ALLOW_OPEN_KEYS=1` for cloud library experiments.

Load an example plugin

With Vite or static server:

```
http://127.0.0.1:3000/plugins/example-coingecko-source.js
```

Manager → Plugins → Load from URL.

Invariants & edge cases

1. **CORS** — Flask must allow the Vite origin; Worker `pickOrigin` allows `:3000` and `:8081`.
2. **Pyodide local** — needs vendor wheels under `public/` (copied to dist on build).
3. **Two endpoints** — topbar endpoint for engine vs cloud storage plugin config can differ (Flask vs Worker).
4. `make` prints the Vite URL `:3000`. Compose serves the built PWA on `:8081`.

Failure modes

Issue	Fix
Engine not ready	Flask down / wrong port
CORS errors	Align origins; see CORS
Plugin 404	File only under `src/plugins/` not `public/plugins/`
Worker scripts 401	Set open keys or mint `pn_` key

See also

- Build and serve
- Worker local
- Plugin examples

CLI

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title: "AXIS CLI" description: "packages/cli — install, doctor, Worker setup (D1/OAuth), secrets, deploy, health — CLI-first operator path."

AXIS CLI

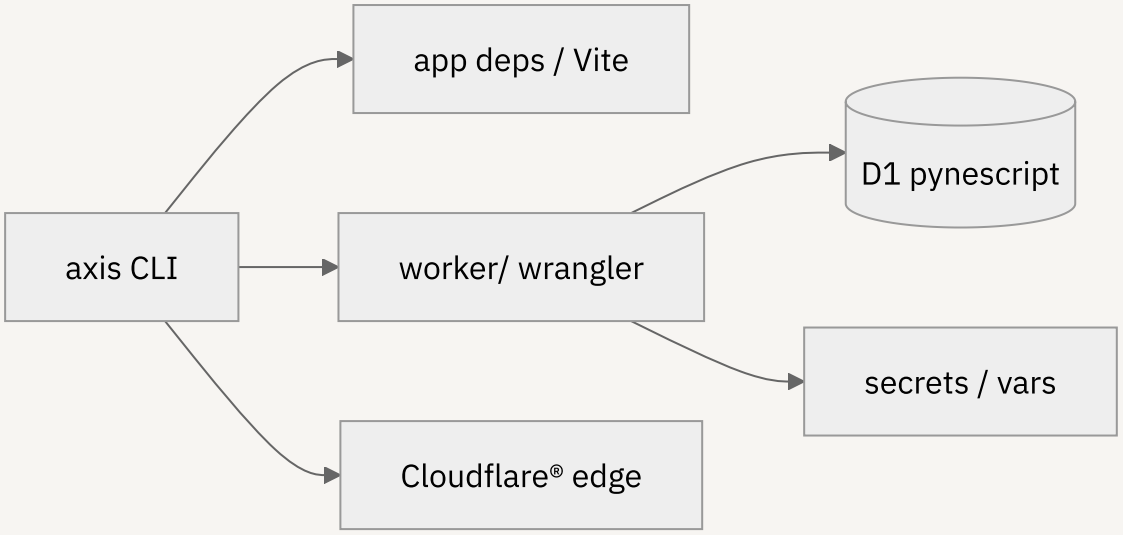
Abstract

The **AXIS CLI** (`@hoox-sh/axis-cli` under `packages/cli`) is the **CLI-first** operator entry for local install, Worker bootstrap, Cloudflare® secrets, deploy, and health checks. It wraps Bun + Wrangler with AXIS-specific paths and defaults (Worker name `worker-axis` , D1 `pynescript` , OAuth device flow).

axis --help listing install, doctor, setup, deploy, health

Requires **Node ≥ 20** to run the published binary (`npm i -g @hoox-sh/axis-cli`) and **Bun ≥ 1.2** for install/dev/build commands that shell out to Bun. Product version follows root `package.json` ; the CLI package (`@hoox-sh/axis-cli`) is versioned independently and publishes on `v*` tags via `.github/workflows/release.yml` .

Conceptual model



Install

```
# npm
npm install -g @hoox-sh/axis-cli
axis --help

# from monorepo root
bun install
cd packages/cli && bun install && bun run build && cd ../..

bun run axis --help
# Make pass-through
make axis ARGS="--help"
# Optional global link
cd packages/cli && bun link && axis --help
```

Repo aliases (bun scripts & Make targets)

`axis` is the primary interface. These aliases exist for working inside the repo checkout without a global install:

Alias	Runs
`bun run axis -- <args>`	`packages/cli/bin/axis.js`
`bun run axis:install`	`axis install`
`bun run axis:doctor`	`axis doctor`
`bun run axis:setup`	`axis setup`
`bun run axis:deploy`	`axis deploy` (default worker)
`bun run axis:health`	`axis health`
`make axis ARGS="doctor --remote"`	same bin with args
`make axis-install` · `axis-doctor` · `axis-setup` · `axis-deploy` · `axis-health`	fixed wrappers

MCP (agents): `axis mcp --key pn_…` (stdio) · `axis mcp config` — see MCP server.

When using `bun run axis setup -- --flag`, keep the `--` so Bun forwards flags to the CLI.

Command surface

Command	Role
`axis install`	`bun install` root + `worker/` + CLI
`axis doctor` [`--remote`]	Toolchain, CF auth, optional live `/health`. Missing `worker/wrangler.toml` is a warning until `axis setup` (expected on a fresh clone).
`axis setup`	Bootstrap: install → ensure toml → local D1
`axis setup --prod`	Production bootstrap: remote D1 schema + `API_KEYS` KV
`axis setup worker`	Copy `wrangler.toml.example` → `wrangler.toml` if missing
`axis setup d1 --local|--remote` [`--create`]	Apply `worker/schemas/scripts.sql` (scripts + version history)
`axis setup kv [--usage]`	Create and bind `API_KEYS` KV (required when D1 is on)
`axis setup oauth --github-client-id …`	Set public OAuth App id in `[vars]` (or `--secret`)
`axis setup artifacts [--namespace axis-scripts] [--skip-check]`	Bind `SCRIPT_ARTIFACTS` in `wrangler.toml` + verify namespace (git-native script storage, closed beta)
`axis keys create [--tier hobby]`	Mint a `pn_…` key via `POST /api/keys` (`ADMIN_TOKEN`)
`axis keys validate --key pn_…`	Validate a key against the Worker
`axis secret put|list|delete`	Wrangler secrets (`ADMIN_TOKEN`, `EXTERNAL_BACKEND`, …)
`axis deploy` / `deploy worker`	Apply remote D1 schema, then deploy Worker `worker-axis`
`axis deploy pages`	Vite build + Pages project
`axis deploy all`	Schema + Worker + Pages
`axis health [--oauth] [--scripts] [--url …]`	Probe `/health`, optional OAuth start, optional `/api/scripts`
`axis mcp [--key pn_…]`	Stdio MCP proxy to Worker `POST /mcp`
`axis mcp config [--stdio]`	Print MCP client JSON (remote URL or stdio spawn)
`axis whoami`	Cloudflare® account
`axis dev`	Vite product UI
`axis dev worker`	Local wrangler Worker
`axis dev desktop`	Tauri desktop shell

Global flags: `--json`, `--quiet`, `-y` / `--yes`.

Installation self-check

Every `axis` command starts with a lightweight, local-only self-check (skipped for `--version` / `--help`, silenced by `--quiet` / `--json`). It never blocks a command — it only prints stderr warnings when something is off:

- **Runtime compliance** vs `engines` (Node ≥ 20, Bun ≥ 1.2)
- **Install context** — npm install, npx cache, or repo checkout
- **Version drift** — a globally installed CLI older than the repo checkout it is operating on (`npm i -g @hoox-sh/axis-cli` to update)

The same check appears as the optional **CLI installation** row in `axis doctor` (and in `doctor --json` as `cli-install`).

axis doctor: required toolchain and wrangler checks passed

Production checklist

```
axis install
axis doctor
axis setup --prod --github-client-id Ov23li…
axis secret put ADMIN_TOKEN   # comments empty [vars] ADMIN_TOKEN, deploys, then sets secret
axis secret put EXTERNAL_BACKEND
axis deploy all
axis keys create              # --admin-token / AXIS_ADMIN_TOKEN / prompt
axis health --scripts --oauth
axis mcp config --key pn_…    # MCP client snippet (see [MCP server](/axis/docs/worker/mcp))
# Paste Worker URL + pn_… into Settings → Script storage
```

Item	Guidance
`GITHUB_OAUTH_CLIENT_ID`	Public OAuth App id; Device Flow enabled on GitHub; **env wins over body `clientId`**
`GITLAB_OAUTH_CLIENT_ID`	Same for GitLab when using git storage
`ADMIN_TOKEN` / `EXTERNAL_BACKEND`	Prefer `axis secret put`. An empty `[vars]` stub collides with secrets (Cloudflare® 10053)
`API_KEYS` KV	`axis setup kv` then `axis deploy worker`. D1 without KV fails closed (`API_KEYS_REQUIRED`)
`ALLOW_OPEN_KEYS`	`"0"` in production
`REQUIRE_RUN_AUTH`	`"1"` to force Bearer on `/api/run` even without KV
Worker script	`worker-axis` (`https://worker.axis.hoox.sh`)
Pages project	`axis` (`https://axis.hoox.sh`)

Env overrides

Variable	Role
`AXIS_ROOT`	Force monorepo root
`AXIS_WORKER_URL`	Default health / deploy probe URL
`AXIS_ADMIN_TOKEN`	Admin token for `axis keys create`
`AXIS_API_KEY`	Default key for `axis keys validate` / `axis mcp`
`CLOUDFLARE_API_TOKEN`	Non-interactive Wrangler auth
`AXIS_CLI_SRC=1`	Load `src/` instead of `dist/`

Internals

Path	Role
`packages/cli/bin/axis.js`	Bin entry (Bun)
`packages/cli/src/commands/*`	Commander handlers (`install` , `doctor` , `setup` , `keys` , `deploy` , `secrets` , `health` , `dev` , `whoami` , `mcp`)
`packages/cli/src/services/wrangler-toml.ts`	Toml vars, D1 id, KV binding upsert
`packages/cli/src/services/health.ts`	`/health` + OAuth probes
Root `package.json` `axis` / `axis:*` scripts	Convenience wrappers
Root `Makefile` `axis` / `axis-*`	Make wrappers

See also: Installation, MCP (agents), MCP server, Cloudflare® deployment, Worker bindings, Worker auth, packages/cli/README.md.

BUILD AND SERVE

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title: "Build and serve" description: "Vite production build, dist layout, axis_pwa_server.py SPA rules, and asset caching."

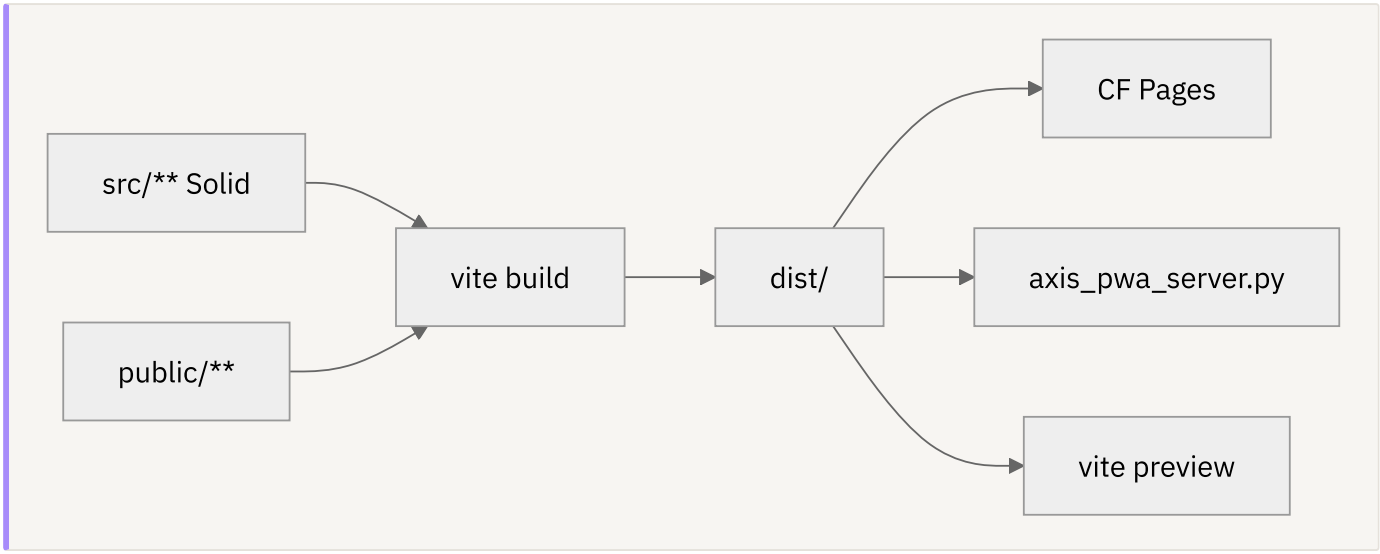
BUILD AND SERVE

Abstract

Production AXIS assets are a **static site** produced by Vite (`bun run build` → `dist`). Serve them with any static host, Cloudflare® Pages, or the repo's `axis_pwa_server.py` SPA-aware server.

Critical requirement: **real binary assets** for Pyodide (`.wasm`, `.whl`, `.zip`) must not be rewritten to `index.html`.

Conceptual model



Interface surface

package.json scripts

```
"dev": "vite",
"build": "vite build",
"preview": "vite preview",
"test": "bun test tests/ worker/tests/",
"test:coverage:gate": "bun run test:coverage && bun scripts/check-coverage.mjs 95",
"test:security": "bun test tests/security/",
"test:e2e:smoke": "playwright test --grep @smoke",
"test:all": "bun run test:coverage:gate && bun run test:security"
```

axis_pwa_server.py

```
bun run build
python axis_pwa_server.py
# HOST / PORT env; default 0.0.0.0:8081 → dist/
```

Behavior (`axis_pwa_server.py`):

Rule	Detail
Root	Serves `dist/`
Cache	`/assets/*` → `immutable` long cache; else `no-cache`
`GET /health` (`/healthz`)	`200` `text/plain` `ok` — never SPA-fallback to `index.html`
`OPTIONS`	`204` with CORS headers (not 501)
`/favicon.ico`	Serves `assets/icon-192.png` when present
SPA fallback	Unknown paths → `/index.html`
Never rewrite	`/health`, `/favicon.ico`, `/assets/`, `/plugins/`, `/vendor/`, `/pyodide/`, or extensions including `.js/.css/.whl/.wasm/.zip/.py/.json…`

This SPA exception list exists because Pyodide + micropip die with opaque `BadZipFile` when HTML is returned for a wheel URL. The engine's `assertZipAsset` detects that class of failure early.

Expected public assets

Path (public → dist)	Purpose
`/plugins/*.js`	Example dynamic plugins
`/vendor/*.whl`	pynescript + antlr wheels
`/pyodide/v0.29.5/**`	Self-hosted Pyodide
`/pyodide/pynescript_runtime.py`	Browser run bridge

Internals

Path	Role
`package.json`	Build scripts
`axis_pwa_server.py`	Threading HTTP SPA server
`src/engines/catalog.ts`	Asset URL expectations
`LEGACY.md`	Product path vs old shell

Makefile note

`make pages-deploy` may historically deploy the wrong tree — **prefer**:

```
bun run build
wrangler pages deploy dist --project-name=axis
```

Build-time flags (Vite)

Env var	Effect
`VITE_APP_VERSION`	Stamps the reported version (CI/Docker); falls back to the repo `VERSION` file
`VITE_MAINTENANCE_MODE`	`1` / `true` / `yes` → the build boots only the maintenance page (`src/ui/MaintenancePage.tsx` : dark screen, flickering HOOX mark, major-release teaser) instead of the chart workspace / editor popout

Both are inlined by Vite at build time — set them in `.env` (local/CI overrides) or export them in the build environment, then rebuild. Existing process env vars outrank `.env` files, which is how e2e forces the maintenance gate off (`playwright.config.ts` → `webServer.env`). The flag is documented in `.env.example` .

Worked example — static desk demo

```
bun install
bun run build
PORT=8081 python axis_pwa_server.py
# another terminal
make run   # Flask :5002
```

Open `http://127.0.0.1:8081` , engine endpoint `http://127.0.0.1:5002` .

Invariants & edge cases

1. **Service workers / offline** — product path uses Vite PWA assets; the legacy root `sw.js` was removed in 2.21.0. Only `public/sw.js` ships.
2. **Relative pyodide index** — resolves against `location.origin` .
3. **Do not hand-edit dist** — regenerate from build.

Failure modes

Symptom	Cause
Blank routes on refresh	Server lacks SPA fallback
Pyodide HTML wheel error	Fallback rewriting `.whl` or missing vendor
Missing plugins	`public/plugins` not copied (check Vite publicDir)

See also

- Local dev
- Cloudflare®
- Engines

DESKTOP

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title: "Desktop (Tauri)" description: "Run and package AXIS as a native desktop app with Tauri 2."

DESKTOP (TAURI)

Abstract

AXIS ships as a browser PWA and as an optional **desktop shell** built with Tauri 2. The Solid/Vite UI is unchanged; Tauri embeds it in a system webview (WebKitGTK on Linux, WKWebView on macOS, WebView2 on Windows).

Commands

From the repo root (after `bun install`):

```
bun run desktop:dev     # Vite :3000 + Tauri window (hot reload)
bun run desktop:build   # Vite production build + native installers
bun run desktop:info    # toolchain / webview diagnostics
```

Script	What it does
`desktop:dev`	Runs `tauri dev` → `beforeDevCommand` starts Vite, opens a native window on `http://127.0.0.1:3000`
`desktop:build`	Runs `tauri build` → `bun run build` then packages `dist/`
`desktop:info`	Prints Rust, system libraries, and webview status

Installers land under `src-tauri/target/release/bundle/` (AppImage, deb, rpm, msi, dmg — depending on host OS and `bundle.targets`).

CI (GitHub Actions)

Every **push to `main`** (and PRs that touch desktop/frontend paths) runs `.github/workflows/desktop.yml`:

Trigger	Result
Push `main`	Matrix build: Linux, macOS arm64, macOS x64, Windows → **workflow artifacts** (14-day retention)
Tag `v*` / `desktop-v*`	Same matrix + **GitHub Release** assets attached to the tag
`workflow_dispatch`	Manual re-run

Artifacts are named `axis-desktop-{linux\|macos-arm64\|macos-x64\|windows}-{sha}`. Concurrency cancels superseded runs on the same ref so each push rebuilds cleanly.

Prerequisites

Always

- Bun (or another JS package manager that can run the scripts)
- Rust (`rustc` / `cargo`) — see `src-tauri/Cargo.toml` `rust-version`
- System webview libraries for your platform (Tauri prerequisites)

Linux (Arch / CachyOS example)

```
sudo pacman -S --needed webkit2gtk-4.1 base-devel curl wget file \
  openssl appmenu-gtk-module libappindicator-gtk3 librsvg patchelf
```

Debian/Ubuntu equivalents use `libwebkit2gtk-4.1-dev` and related `-dev` packages (see the Tauri docs for the current list).

Engine backends

Desktop AXIS is the same product as the PWA. For Pine evaluation you still need one of:

Engine	Notes
Local **pyne** Pro API	Clone hoox-sh/pyne as `../pyne` (`make run` → `:5002` ; Vite proxies `/run` in dev). Local checkouts are sometimes still named `pynescript` .
Cloudflare® Worker	Configure worker URL in Manager
Pyodide (in-webview)	Fully offline path; largest first load

Layout

Path	Role
`src-tauri/`	Rust host, `tauri.conf.json` , icons, capabilities
`src-tauri/tauri.conf.json`	App id `sh.hoox.axis` , window size, build hooks
`src/`	Unchanged Solid product UI
`dist/`	Vite output consumed by `frontendDist`

Behaviour notes

- **Service worker** is skipped in the Tauri shell (`src/pwa/register-sw.ts`). Offline install is a PWA concern; the desktop app is already installed as a native binary.
- **Window** defaults: 1440×900, min 960×640, centered.
- **CSP** is left open (`null`) so external venues, Pyodide assets, and worker APIs work the same as in the browser. Tighten later if you ship a locked-down build.

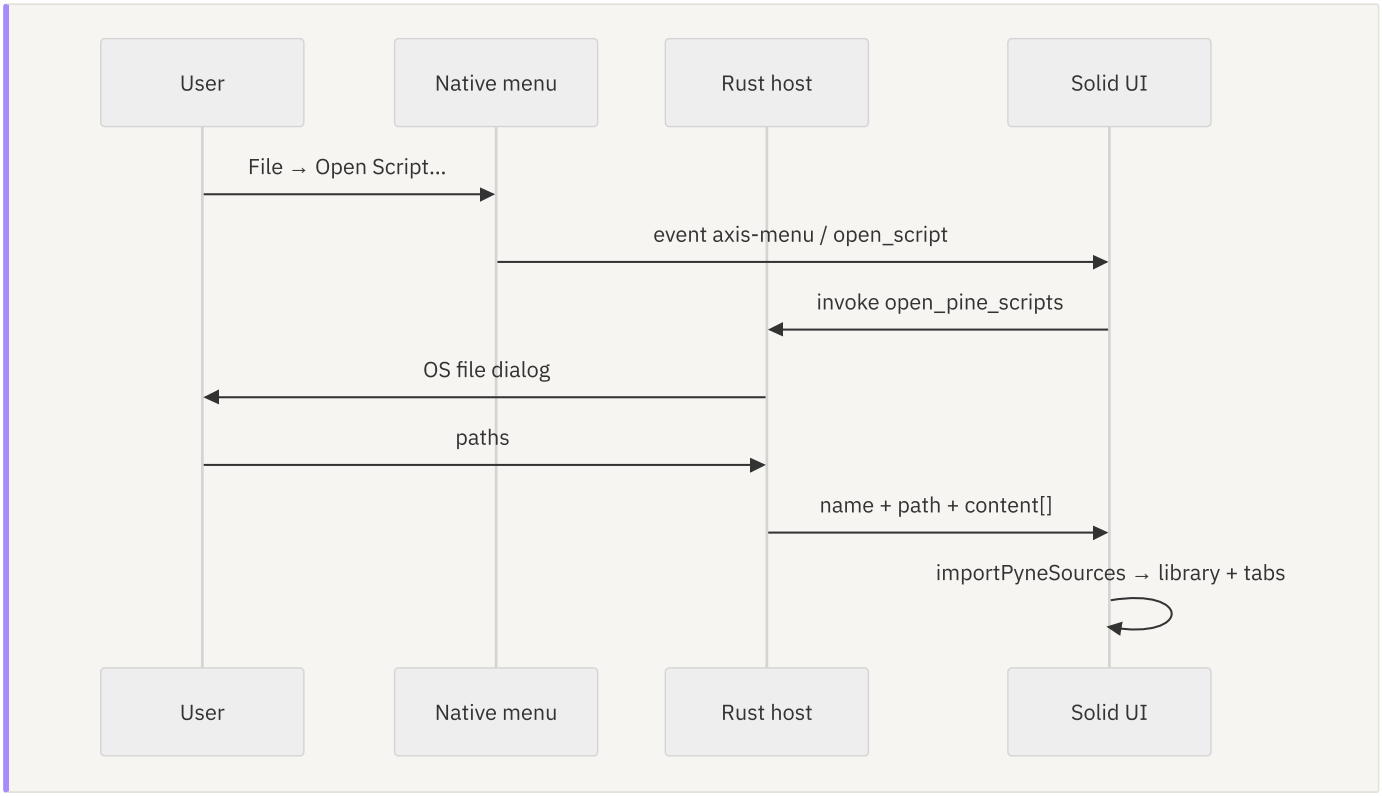
- Icons are generated from `public/assets/icon-512.png` via `bunx tauri icon`.

Native menu & open from disk

The Tauri host builds a native app menu:

Menu	Item	Action
File	**Open Script…** (`⌘/Ctrl+O`)	System multi-file dialog → library + editor tabs
File	**Quit AXIS**	Quit (platform predefined)
Help	**About AXIS**	Info dialog (version / license / site)

Flow

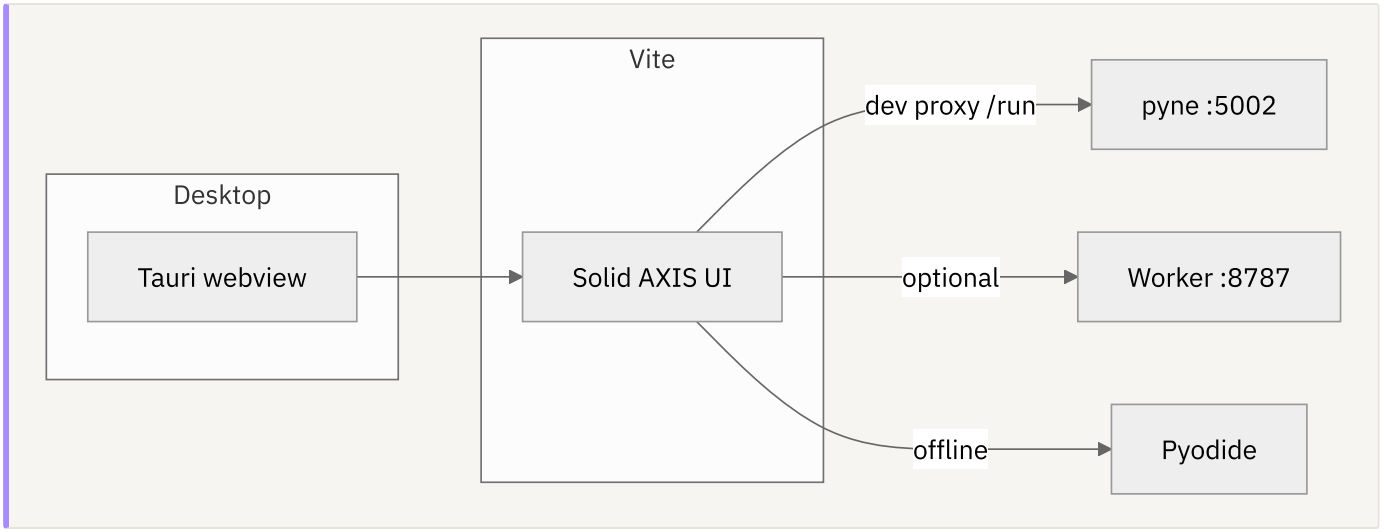


Path	Role
`src-tauri/src/lib.rs`	Menu, `open_pine_scripts` command, dialog + disk read
`src/desktop/`	`isTauriShell` , menu listen, open + About wiring
`src/storage/import-pyne-open.ts`	Shared status/logs/editor open after import
`src/storage/import-pyne-files.ts`	`importPyneSources` (text) + `importPyneFiles` (browser `File`)

Accepted extensions match the PWA drop path: `.pyne` , `.pine` , `.pinescript` , `.pinev5` , `.pinev6` . Per-file size cap on the host: **8 MiB**.

Drag-and-drop of the same extensions still works in the desktop webview.

Conceptual model



CLOUDFLARE

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title: "Cloudflare® deployment" description: "Pages + Worker for AXIS: Pages axis.hoox.sh (project axis), Worker worker.axis.hoox.sh, CLI deploy, bindings, security checklist."

CLOUDFLARE® DEPLOYMENT

Abstract

Production AXIS on Cloudflare® is a **split deploy**:

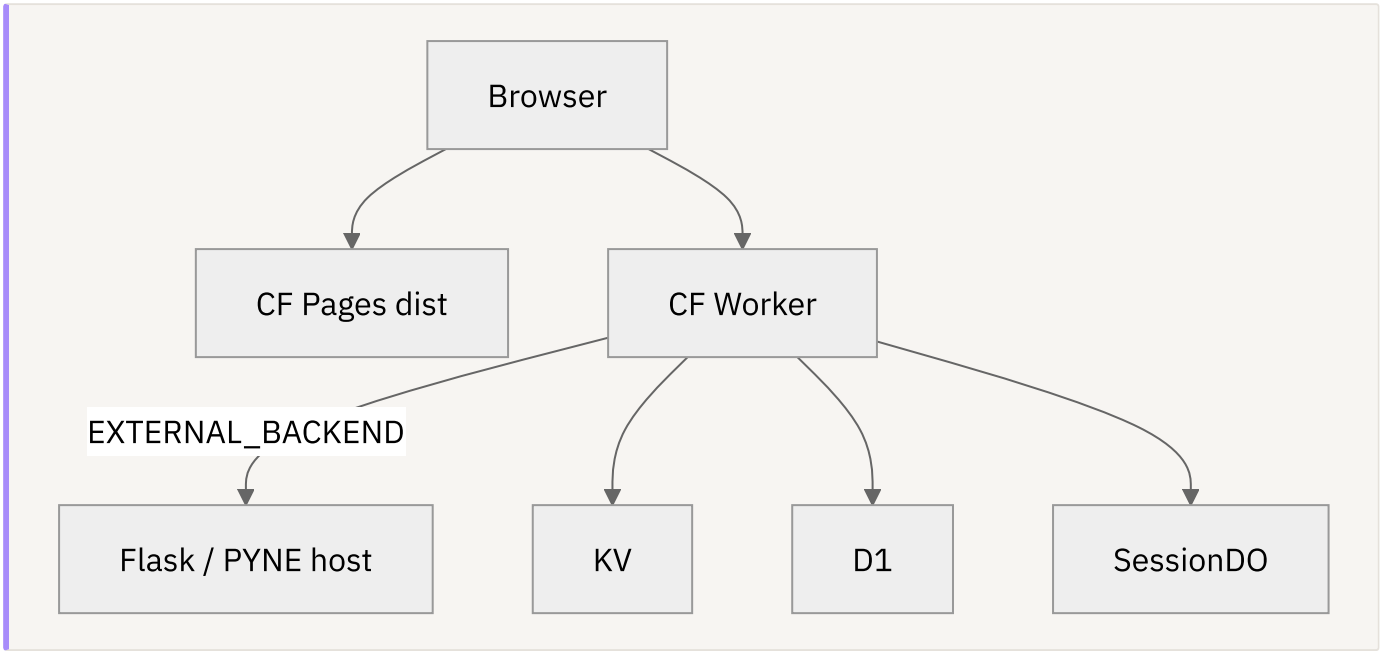
Piece	Role
Pages	Static PWA (`dist/`)
Worker	JSON API + WebSocket (`worker/`)

Pages: project `axis` , canonical `https://axis.hoox.sh` . **Worker:** script `worker-axis` at `https://worker.axis.hoox.sh` .

Evaluation truth: Worker `/api/run` **proxies to** `EXTERNAL_BACKEND` unless the gated Pyodide path is fully implemented and enabled. You still need a Python host (VPS Flask, container, etc.) for real Pine Script™ fidelity today.

Prefer the **AXIS CLI** for setup and deploy (`axis deploy`).

Conceptual model



Deploy procedure

1. Provision bindings (once)

See Worker bindings:

- KV `API_KEYS`, `USAGE`
- D1 `pynescript` + schema
- Optional R2, Durable Objects

2. Configure secrets / vars

Production checklist:

Setting	Production guidance
`EXTERNAL_BACKEND`	Public HTTPS URL of Flask/PYNE API (`axis secret put`)
`ALLOWED_ORIGIN`	Extra exact origins (comma-separated); product hosts + `*.axis.pages.dev` are built-in — CORS
`ADMIN_TOKEN`	Secret, strong
`API_KEYS` KV	**Bind in prod** — D1 without KV fails closed (`API_KEYS_REQUIRED`)
`ALLOW_OPEN_KEYS`	`"0"` or unset
`REQUIRE_RUN_AUTH`	`"1"` optional force Bearer on `/api/run`
`GITHUB_OAUTH_CLIENT_ID`	Env preferred; never rely on body `clientId` alone
`PYODIDE_IN_WORKER`	`"disabled"` until wheel pipeline ready

3. Deploy Worker

```
# Preferred: AXIS CLI
axis deploy
# or: make axis-deploy

# Schema first (remote)
axis setup d1 --remote

# Equivalent raw wrangler
cd worker
bun install
bun run deploy
```

4. Build & deploy Pages

```
bun run axis deploy pages
# or
bun run build
bunx wrangler pages deploy dist --project-name=axis
# Make: make pages-deploy
```

5. Point the PWA

- Engine endpoint: Worker URL (if path-mapped to `/run`) **or** Flask URL directly
- Cloud storage endpoint: Worker origin
- Ensure CORS allows Pages origin

Internals

Path	Role
`worker/wrangler.toml` / `.example`	Worker name + bindings
`worker/README.md`	Ops narrative
`worker/package.json`	deploy scripts
`packages/cli/`	`axis deploy` / `axis health`
Makefile `axis-deploy` / `pages-deploy` / `worker-deploy`	Convenience targets

Health check

```
axis health
# or
curl -sS https://<worker>/health
# service: worker-axis
```

Invariants & edge cases

1. **Pages ≠ Worker host** — CORS required unless same-origin routing via custom domain routes.
2. **Localhost EXTERNAL_BACKEND** is useless from the edge.
3. **DO migrations** apply on deploy when uncommented.
4. **Observability** enabled in wrangler for log tails (`wrangler tail`).

Failure modes

Issue	Fix
503 NO_BACKEND	Set reachable EXTERNAL_BACKEND
CORS blocked	ALLOWED_ORIGIN mismatch
SPA HTML for wheels	Pages must serve `/vendor` and `/pyodide` as static files
Stream NO_DO	Enable SessionDO bindings

See also

- Worker
- CORS
- VPS demo (backend co-host)

VPS DEMO

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title: "VPS demo topology" description: "Single-box demo: static AXIS dist + Flask Pro API (+ optional reverse proxy) without Cloudflare®."

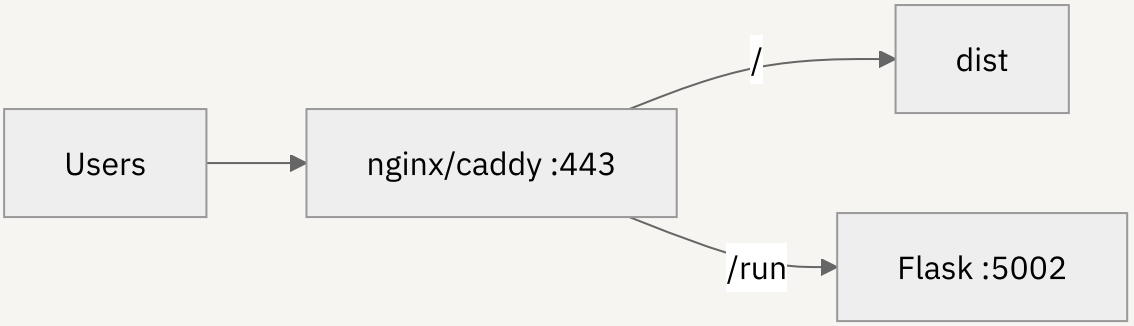
VPS DEMO TOPOLOGY

Abstract

A **VPS demo** is the lowest-surprise production-like setup for Pine evaluation: one machine runs Flask (PYNE) and serves the static AXIS build. Optional nginx/caddy terminates TLS and reverse-proxies `/run` same-origin to avoid CORS.

No Worker required. Cloud storage and DO streams will not work without the Worker.

Conceptual model



Interface surface

Processes

Process	Command	Port
Build (CI or once)	`bun run build`	—
Static	`PORT=8081 python axis_pwa_server.py`	8081
API	`make run` / `python -m backend.app`	5002

Same-origin proxy sketch (Caddy)

```
axis.example.com {
  handle /run* {
    reverse_proxy 127.0.0.1:5002
  }
  handle /optimize* {
    reverse_proxy 127.0.0.1:5002
  }
  handle {
    reverse_proxy 127.0.0.1:8081
  }
}
```

Then PWA endpoint = `https://axis.example.com` and the **server** engine posts to `/run` on the same host.

Production (split: Pages PWA + Hetzner API)

- **PWA** — Cloudflare® Pages custom domain `https://axis.hoox.sh`
- **PYNE Pro API** — Hetzner VPS `https://pynescript.online` (Cloudflare® orange-cloud → origin `204.168.138.51`). nginx `:443` reverse-proxies `/run`, `/optimize`, `/ws/`, `/health`, `/lsp/`, `/datafeed/`, … to gunicorn `127.0.0.1:5002`
- PWA **Backend URL** = `https://pynescript.online` (cross-origin; product CORS regex)
- Cloudflare® Pages previews (`*.axis.pages.dev`) use the same Backend URL — pyne must allow those Origins on `/health` and `/run` (always-on product regex; see CORS)

SSH alias: `ssh pynescript` (`204.168.138.51`). The previous combined VPS (`ssh axis` / `162.254.38.194`) stays until decommission.

Update deploy (from a machine that can SSH to the VPS):

```
# 1. Ship latest main
git push origin main

# 2. On VPS (paths may vary — often /root/axis + rsync dist → PWA WorkingDirectory)
# Pro API (from pynescript checkout)
./scripts/deploy_vps.sh
# or:
ssh pynescript 'cd /root/pynescript && git pull --ff-only && systemctl restart pynescript-api'
# PWA: `bun run build` + wrangler pages deploy (axis.hoox.sh)
```

Local operator CLI for the **Worker** edge (separate from VPS static): `axis deploy` — see AXIS CLI.

Hardened network (UFW) + health checks

Production VPS pattern (verified on `pynescript.online`):

Surface	Bind / allow	Health URL
UFW	`deny incoming` default; **allow 22/tcp, 80/tcp, 443/tcp**	—
nginx	`0.0.0.0:443` TLS (+ `:80` ACME)	`https://pynescript.online/health` → proxy `127.0.0.1:5002/health`
Pro API (gunicorn)	**`127.0.0.1:5002` only**	loopback; **not** public
PWA static	Cloudflare® Pages (`axis.hoox.sh`)	not on this VPS
fail2ban	`sshd` jail	—

Do not open UFW for `5002`, `80`, or `8787` on a public VPS. Direct probes to `http://<vps-ip>:5002/health` **must time out** if hardening is correct.

Workers Manager browser probes:

- On **product API host** (`pynescript.online`), pyne Pro is probed via **same-origin** `/health` (nginx), not `http://127.0.0.1:5002` (that would hit the *client* PC and always look “down”).
- From **CF Pages** (`axis.hoox.sh` / `*.axis.pages.dev`) catalog `publicEndpoint` (`https://pynescript.online`) is used — never client loopback.
- CF Worker health stays on `https://worker.axis.hoox.sh/health` (edge; independent of VPS UFW).

Minimal without TLS

- Static `:8081`, Flask `:5002`
- Prefer a hostname over raw IP when the page is HTTPS
- Open firewall; accept CORS from static origin (configure Flask accordingly)

Internals

Concern	Notes
Pyodide offline	Ship full `dist` including `vendor` + `pyodide`
Process supervisor	systemd units for flask + axis_pwa_server
Updates	Rebuild dist artifact; restart static only
Secrets	Flask admin / pro keys separate from Worker

Worked example — systemd sketch

```
# /etc/systemd/system/axis-pwa.service
[Service]
WorkingDirectory=/opt/axis
Environment=PORT=8081
ExecStart=/usr/bin/python3 axis_pwa_server.py
Restart=on-failure
```

Pair with an existing `backend` unit for Flask.

Invariants & edge cases

1. **CPU/RAM** — Pyodide is browser-side; VPS load is Flask evaluate concurrency.
2. **Do not expose open admin tokens**.
3. **Disk** — dist + wheels are tens of MB.
4. Hybrid: VPS Flask as `EXTERNAL_BACKEND` for a CF Worker still works if VPS has a public URL.

Failure modes

Issue	Fix
Mixed content	HTTPS page calling HTTP Flask blocked
CORS	Prefer same-origin proxy
502 on /run	Flask crashed; check journalctl

See also

- Build and serve
- Cloudflare®
- CORS

CORS AND ORIGINS

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title: "CORS and origins" description: "Worker pickOrigin: local-dev, product hosts, project-scoped Pages previews, ALLOWED_ORIGIN list, Flask constraints."

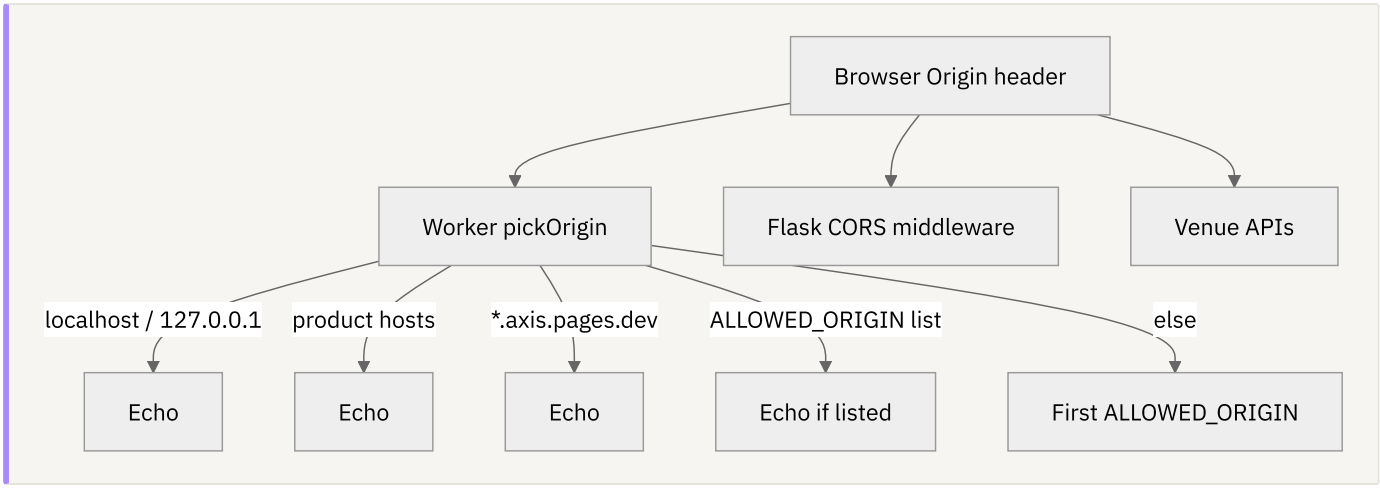
CORS AND ORIGINS

Abstract

AXIS is a browser product that talks to **cross-origin** backends (Flask, Worker, venues). CORS misconfiguration is the #1 local-deploy footgun after missing wheels.

The Worker implements an explicit origin picker (`pickOrigin` in `worker/src/index.ts`); Flask and venue APIs have their own rules.

Conceptual model



Worker behavior

Resolution order for `Access-Control-Allow-Origin`:

1. Local-dev (always echoed)

Any `http://` or `https://` origin on **`localhost`** or **`127.0.0.1`**, optional port (e.g. Vite `:3000`, `axis_pwa_server` `:8081`, Playwright ephemeral ports).

Not allowed (and not needed): `http://0.0.0.0:…` — browsers do not use `0.0.0.0` as a page origin; binding with `HOST=0.0.0.0` only means "listen on all interfaces."

Localhost/127 allowlisting is **safe for CORS**: only pages actually served from those hosts present that `Origin`. A public site cannot forge `Origin: http://localhost:3000` in a real browser.

2. Product hosts (2.0.1+)

Exact product / HOOX / legacy PYNE hosts under:

- `*.hoox.sh` / `hoox.sh`
- `*.pynescript.online` / `pynescript.online`

Examples: `https://axis.hoox.sh` (canonical Pages), `https://hoox.sh`, `https://pynescript.online`.

3. Project-scoped Cloudflare® Pages (not open `*.pages.dev`)

Canonical Pages is **`https://axis.hoox.sh`** (product host above). Preview hashes on **`*.axis.pages.dev`** (and the apex `axis.pages.dev`) are also echoed. Legacy **`*.pynescript-axis.pages.dev`** still matches. Arbitrary third-party Pages projects (`evil.pages.dev`, other apps) fall through to the allowlist fallback — they are **not** reflected.

4. `ALLOWED_ORIGIN` allowlist

Comma-separated exact origins in `env.ALLOWED_ORIGIN` (e.g. `https://a.example,https://b.example`). If the request Origin is listed, echo it; otherwise return the **first** entry (default `https://pynescript.online`).

CORS headers applied:

```
Access-Control-Allow-Origin: <picked>
Access-Control-Allow-Methods: GET, POST, PUT, DELETE, OPTIONS
Access-Control-Allow-Headers: Content-Type, Authorization, X-Admin-Token, If-Match
Access-Control-Max-Age: 86400
Vary: Origin
```

`OPTIONS` → `204` preflight.

Production implication

For a custom domain already under product regex, no extra var is required. For one-off preview hosts outside the project, list them:

```
ALLOWED_ORIGIN = "https://my-preview.example.com,https://axis.example.com"
```

Flask / server engine

The **server** engine posts from the browser to the configured endpoint. Flask Pro API must:

1. Answer `OPTIONS` preflight if cross-origin.
2. Reflect or allow the PWA origin.
3. Accept `Content-Type: application/json`.

Pyne always appends the same product Origin regex as Worker `pickOrigin` (including `*.axis.pages.dev`) even when systemd `ALLOWED_ORIGINS` is a short list. `GET /health` (AXIS Settings probe) and `POST /run` are **free CORS paths** — they echo any `Origin` so a Pages preview can reach `https://axis.hoox.sh` without a per-preview allowlist entry.

Same-origin reverse proxy (VPS demo) eliminates CORS when the PWA and Pro API share `https://axis.hoox.sh`. Cross-origin Pages → VPS still needs those headers (they are now default).

Venue sources/streams

Public Binance/OKX/etc. REST must send CORS headers usable by browsers. Firefox often reports **“CORS request did not succeed / status (null)”** when the TCP/TLS connection fails entirely (geo block, firewall, extension) — that is **not** a missing `Access-Control-Allow-Origin` header.

AXIS resilience for Binance:

1. Direct hosts: `api.binance.com` then `data-api.binance.vision` (`src/data/binance-http.ts`).
2. Worker allowlisted proxy: `GET /api/market/binance/klines|ticker/24hr|exchangeInfo` on the AXIS Worker (`worker/src/market.ts`).
3. Live WS rotates `:9443` → `:443` → `data-stream.binance.vision` (port 9443 is frequently firewalled).
4. Offline last resort: `mock-walk` / `mock-poll`, or warm bars-cache.

Prefer the DO relay (`cf-do` stream / `/api/stream`) when many tabs share one upstream WS — still requires the Worker edge to reach Binance.

AXIS resilience for MEXC:

1. Worker allowlisted proxy first: `GET /api/market/mexc/{klines,ticker/24hr,exchangeInfo}` (`src/data/mexc-http.ts`). Public `api.mexc.com` does **not** send `Access-Control-Allow-Origin`.

2. Direct `api.mexc.com` only as last-ditch fallback (offline lab, tests, or Worker 5xx). Worker 4xx is not retried against the CORS-blocked host.

Dynamic plugins

- **Module import** of plugin JS: needs CORS on the **script origin**. Prefer same-origin `/plugins/*` (PYNE Agent ships as `/plugins/axis-pine-agent.js`).
- **Remote** modules (e.g. legacy `pyne-agent-worker…/plugin/axis-pine-agent.js`) need ACAO + CSP `script-src` allow when a CSP is present.
- **fetch inside plugin**: subject to third-party CORS independently (agent API still uses the pyne-agent Worker origin).

Invariants & edge cases

1. **Credentials** — Worker CORS does not set `Allow-Credentials: true`; use Bearer headers, not cookies.
2. **Multiple prod domains** — product regex + comma-separated `ALLOWED_ORIGIN` cover multi-host; unknown sites never get open reflection.
3. **Vite vs 8081** — both covered for Worker; Flask config must match whichever you use.
4. **Health checks** from curl omit Origin — still work; browser calls need correct ACAO.
5. **No open `*.pages.dev`** — only the Pages project `axis` previews (`*.axis.pages.dev`) are product-scoped.

Failure modes

Browser console	Meaning
blocked by CORS policy	Origin not allowlisted
preflight 404	Server lacks OPTIONS
No ACAO on error body	Some error paths forgot headers (Worker try/catch mostly covered)

See also

- Worker bindings
- Local dev
- Dynamic loader

CI AND TESTING

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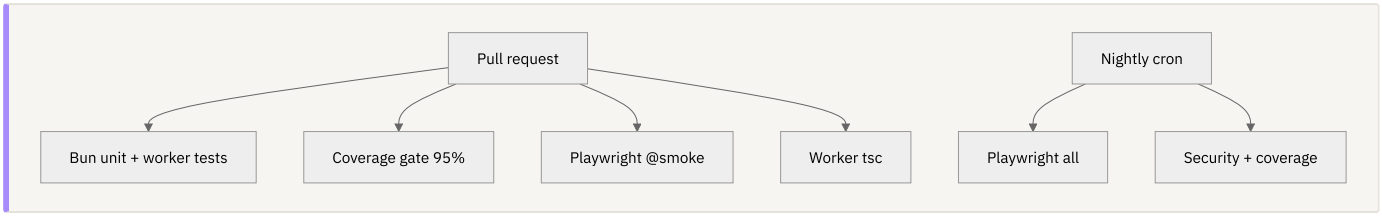
title: "CI and testing" description: "GitHub Actions for AXIS: unit coverage gate, Playwright smoke, worker typecheck, nightly full e2e."

CI AND TESTING

Abstract

AXIS quality gates live primarily under the repo root (Bun tests) and `worker/tests/`, orchestrated by GitHub Actions (`.github/workflows/ci.yml`, `axis-nightly.yml`). Deep test conventions are documented in Testing reference and `TESTING.md`.

Conceptual model



Interface surface — local commands

```
bun run test                # tests/ + worker/tests/
bun run test:unit
bun run test:coverage:gate  # lcov + scripts/check-coverage.mjs 95
bun run test:security
bun run test:e2e:smoke
bun run test:e2e:critical
bun run test:e2e
bun run test:all            # gate + security
```

Worker:

```
cd worker
npm run typecheck
```

Repo Makefile:

```
bun run test:unit
make worker-typecheck
```

CI workflows

ci.yml (excerpted jobs)

Job (names vary)	What
Frontend tests / coverage	Bun install, coverage, artifact `axis-coverage-lcov`
`axis-e2e`	Playwright `@smoke` on PR
Worker typecheck	`worker` tsc
Related	Python suite remains for pynescript core

axis-nightly.yml

Job	What
Playwright full	all e2e specs
Security + coverage gate	`test:coverage:gate` + `test:security`

Schedule: `0 4 * * *` UTC + `workflow_dispatch`.

Internals

Path	Role
`TESTING.md`	Authoritative testing guide
`scripts/check-coverage.mjs`	Scoped line gate
`playwright.config.ts`	webServer build+preview `:4173`
`e2e/`	Specs with `@smoke` / `@critical` tags
`worker/tests/`	auth, keys, scripts, runtime, http, proxy-router, rate-limit
`.github/workflows/ci.yml`	PR gates
`.github/workflows/axis-nightly.yml`	Nightly

Coverage policy (summary)

- **Gate:** 95% lines on a **scoped** core (plugins, storage minus idb, store, results, sources, streams, data, selected chart helpers, worker auth/keys/runtime/scripts).
- **Excluded from gate:** heavy chart UI, pyodide boot, legacy JS, Solid `.tsx` surfaces (unit-tested elsewhere or deferred).

E2E design

- Builds production preview; baseURL `http://127.0.0.1:4173`.
- Mocks `/run` and Binance where possible so smoke does not need live Pro API.
- Selectors use `data-testid` (`axis-btn-load`, `axis-manager`, …).

Storybook

`bun run storybook` (`:6006`) serves `src/stories/` in isolation — visual review without Flask/Worker. Not a CI gate today.

Invariants & edge cases

1. **No real network in unit tests** — mock `fetch` / WebSocket.
2. **Import `./setup` first** when touching store/plugins.
3. **CI Bun version** pinned in workflows (e.g. 1.3).
4. **Playwright browsers** installed with `--with-deps` on CI.

Failure modes

CI red	Likely
Coverage gate	New untested code in scoped paths
Smoke timeout	Build slow; webServer 180s
Worker tsc	Env typing / DO exports
Flaky e2e	Missing testid; race without mock

See also

- Testing reference
- Local dev

OKF

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title: "OKF bundle" description: "Compiled Open Knowledge Format cache of the AXIS repo, refreshed on every commit."

OKF BUNDLE

Abstract

`okf/` is an Open Knowledge Format v0.2 bundle. It is the compiled map of this repo: one concept per source directory and per doc, with imports already linked. Agents read `okf/index.md` and then one concept instead of searching the tree.

Refresh

The pre-commit hook (`.githooks/` , installed by `bun install`) runs the pipeline against the git index and stages the generated bundle into the commit. Unstaged edits, including unstaged edits to curated playbooks, are not part of that commit's bundle.

```
bun run okf:enrich    # draft, link, and lint from the working tree
bun run okf:check     # fail if the committed bundle would change
bun run okf:lint      # conformance errors and producer warnings
bun run okf:query chart drawings
bun run okf:context src/chart
```

A directory path resolves to that concept. A file path resolves to the deepest directory concept that contains it. Query terms match path segments and the file names listed on a concept, so `ui` does not match `builtins`, and `drawings.ts` finds the directory that contains it. `AGENTS.md` is gitignored and is not part of this bundle. The shared reading guide is `okf/playbooks/read-this-first.md`.

`OKF_SKIP=1` skips the hook refresh. CI runs `okf:check`, so a commit that bypasses the hook still cannot merge a stale bundle.

Generated concepts (`okf_lock: generated`) are replaced when their sources change. Curated playbooks (`okf_lock: human`) are kept. Put hand-written knowledge in `okf/playbooks/` with that lock. The draft records exports, imports, and the first non-license comment. It does not call a model.

Layout

Path	Role
`okf/index.md`	Bundle root. Declares `okf_version: "0.2"`.
`okf/log.md`	Newest-first compile history.
`okf/code/`	One concept per source directory.
`okf/docs/`	Outlines of `docs/`.
`okf/playbooks/`	Curated reading guide and product map.
`scripts/okf/`	Compiler, linter, and query CLI.